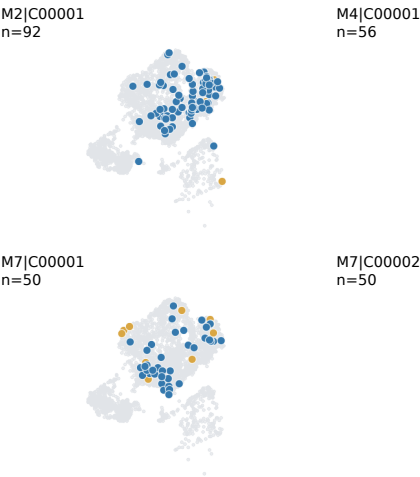


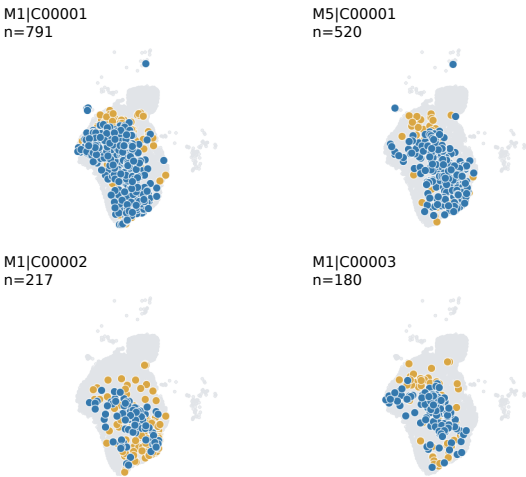
Supplementary Figure 2 | Model-antigen clone relationships and orthogonal GC labels

A NP-OVA clones spanning division gates



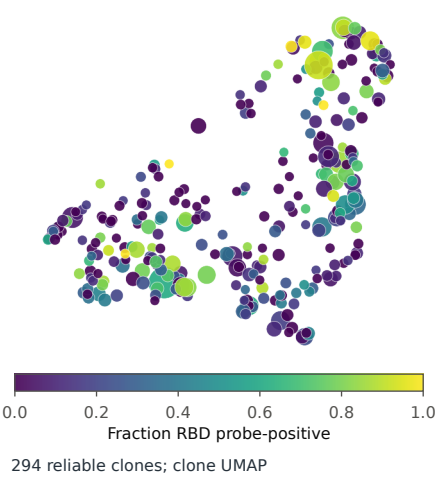
Each inset shows one BCR-defined clone on the same cell UMAP.

B RBD clones spanning division gates

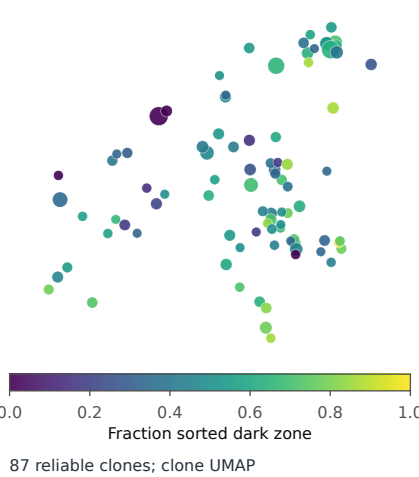


Each inset shows one BCR-defined clone on the same cell UMAP.

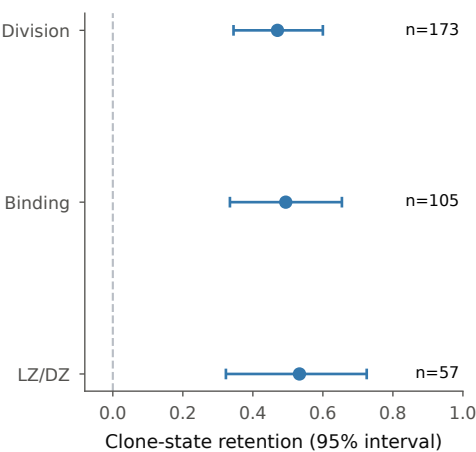
C RBD protein: antigen binding



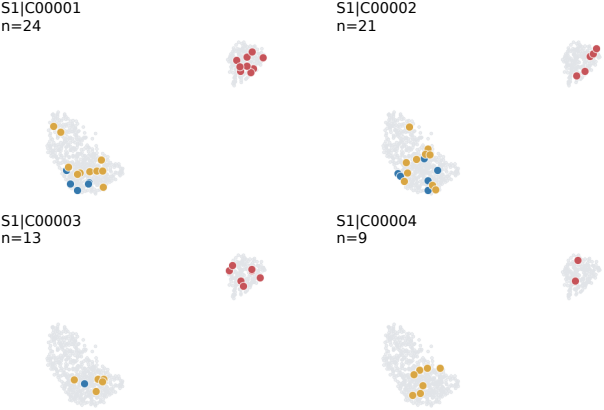
D RBD mRNA: DZ occupancy



E State across GC gates



F Directly measured NP-OVA output compartments



Each inset shows one BCR-defined clone on the same cell UMAP.

G Library-stratified reporter associations

